NANO SENTINEL

INTELLIGENT WAFER INSPECTION REPORT

Inspection field	Result
Inspection ID	NS-000001
Timestamp (UTC)	2026-10-12T13:35:07.289113+00:00
Model	MobileNetV2 V2
Predicted defect class	Edge-Loc
Model confidence	96.52%
Inspection status	MANUAL REVIEW

1. Inspection Interpretation

The MobileNetV2 V2 model predicted Edge-Loc with a confidence score of 96.52%. The reference label for this dataset example is Edge-Ring. Since the prediction and reference label disagree, this example requires manual review. A high confidence score does not guarantee that a prediction is correct.

2. Grad-CAM Explainability

Grad-CAM highlights image regions that influence the model's prediction. The heatmap is a qualitative explanation of model behaviour; it does not prove the physical location or cause of a manufacturing defect.

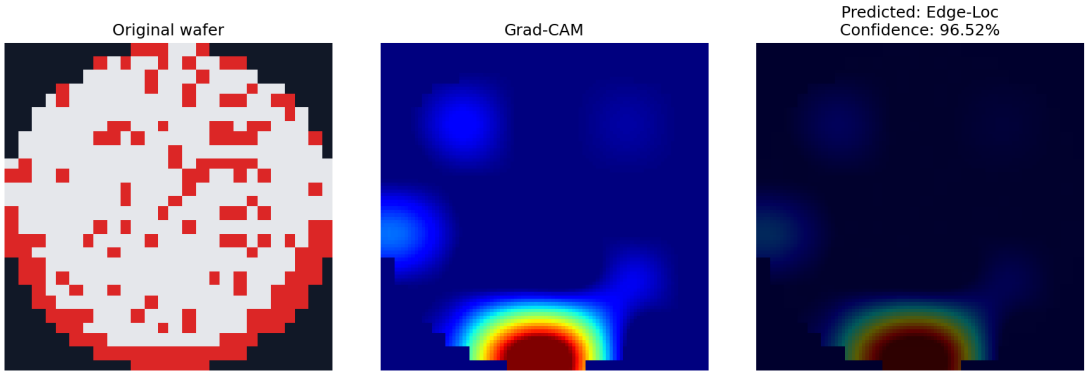


Figure 1. Saved wafer inspection visualization and Grad-CAM overlay.

3. Wafer Defect Statistics

Metric	Value	Explanation
Total die pixels	533	Pixels included in the encoded die area.
Defective die pixels	154	Pixels marked defective in the wafer map.
Defect ratio	28.89%	Defective pixels divided by total die pixels.
Provisional category	HIGH	Not validated against manufacturing standards.

Interpretation: the 28.89% figure is a ratio of encoded map pixels, not a validated measurement of physical or economic severity. The HIGH category is provisional.

4. Error Analysis

Observed error: the reference label is Edge-Ring, but the model predicted Edge-Loc. The prediction has 96.52% confidence, making this a high-confidence misclassification in this labelled example.

Top model outputs: Edge-Loc: 96.52%; Loc: 2.32%; none: 0.87%.

Possible factors to investigate: similarity between Edge-Loc and Edge-Ring patterns, imbalanced training classes, differences between training and evaluation data, and inconsistent preprocessing. These are hypotheses, not confirmed causes of this error.

5. Recommended Improvements

• Class-wise evaluation: review per-class precision, recall, F1-score and confusion matrices instead of relying on accuracy alone.

• Preprocessing consistency: verify that training, validation, Grad-CAM and deployment use identical resizing, padding, channel handling and normalization.

• Minority-class coverage: examine the quantity and diversity of rare-class examples. Test class-aware sampling or augmentation using training data only.

• Confidence calibration: measure whether predicted confidence matches actual correctness. Set review thresholds using validation data; do not equate high confidence with correctness.

• Explainability validation: compare Grad-CAM maps with known wafer patterns and domain-expert judgement.

• Severity validation: validate map-based thresholds against agreed manufacturing criteria before operational use.

6. Decision and Next Action

Current status: MANUAL REVIEW. Review the original wafer map and Grad-CAM visualization, verify the label, and document the review outcome before considering corrective action. For ordinary uploads, the true label will not be known, so predictions must be presented as model estimates rather than confirmed diagnoses.

7. Limitations

This report describes one labelled dataset example, not a complete model benchmark. The defect ratio is a map statistic, not a calibrated physical severity score. The reference label is available only because this is a labelled example. Grad-CAM is qualitative, and the suggested improvements require testing and validation before production use.